①

Find a basis for the nullspace of

$$A = \begin{pmatrix} 1 & 4 & 5 & 2 \\ 2 & 1 & 3 & 0 \\ -1 & 3 & 2 & 2 \end{pmatrix}$$

Solⁿ: The nullspace of A is the solution space of the homogeneous system.

The augmented matrix for the system is

$$\left(\begin{array}{cccc|c} 1 & 4 & 5 & 2 & 0 \\ 2 & 1 & 3 & 0 & 0 \\ -1 & 3 & 2 & 2 & 0 \end{array}\right) \begin{array}{l} R_2' = R_2 - 2R_1 \\ R_3' = R_3 + R_1 \end{array}$$

$$\left(\begin{array}{cccc|c} 1 & 4 & 5 & 2 & 0 \\ 0 & -7 & -7 & -4 & 0 \\ 0 & 7 & 7 & 4 & 0 \end{array}\right) R_3' = R_3 + R_2$$

$$\left(\begin{array}{cccc|c} 1 & 4 & 5 & 2 & 0 \\ 0 & -7 & -7 & -4 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right) R_2' = \frac{R_2}{-7}$$

$$\left(\begin{array}{cccc|c} 1 & 4 & 5 & 2 & 0 \\ 0 & 1 & 1 & 4/7 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

$$x_1 + 4x_2 + 5x_3 + 2x_4 = 0$$

$$x_2 + x_3 + \frac{4}{7}x_4 = 0$$

Let $x_3 = s$ and $x_4 = t$

$$x_2 = -s - \frac{4}{7}t$$

$$x_1 = -4\left(-s - \frac{4}{7}t\right) - 5s - 2t$$

$$= 4s + \frac{16}{7}t - 5s - 2t$$

$$= \cancel{\frac{28s + 16t \ldots}{7}}$$

$$= -s + \frac{2t}{7}$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} -s + \frac{2}{7}t \\ -s - \frac{4}{7}t \\ s \\ t \end{pmatrix}$$

$$= s\begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} \frac{2}{7} \\ -\frac{4}{7} \\ 0 \\ 1 \end{pmatrix}$$

Nullspace of A is, $N(A) = \left\{ s\begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} \frac{2}{7} \\ -\frac{4}{7} \\ 0 \\ 1 \end{pmatrix} \right\}$

The basis for $N(A)$ is $= \left\{ \begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{2}{7} \\ -\frac{4}{7} \\ 0 \\ 1 \end{pmatrix} \right\}$

Nullity = 2

(2) Find the rank and nullity of the matrix

$$A = \begin{pmatrix} 1 & 4 & 5 & 6 & 9 \\ 3 & -2 & 1 & 4 & -1 \\ -1 & 0 & -1 & -2 & -1 \\ 2 & 3 & 5 & 7 & 8 \end{pmatrix}$$

Soln: Reduce to row echelon form.

$$\begin{pmatrix} 1 & 4 & 5 & 6 & 9 \\ 3 & -2 & 1 & 4 & -1 \\ -1 & 0 & -1 & -2 & -1 \\ 2 & 3 & 5 & 7 & 8 \end{pmatrix} \begin{matrix} \\ R_2' = R_2 - 3R_1 \\ R_3' = R_3 + R_1 \\ R_4' = R_4 - 2R_1 \end{matrix}$$

$$\begin{pmatrix} 1 & 4 & 5 & 6 & 9 \\ 0 & -14 & -14 & -14 & -28 \\ 0 & 4 & 4 & 4 & 8 \\ 0 & -5 & -5 & -5 & -10 \end{pmatrix} \begin{matrix} \\ R_2' = \frac{R_2}{-14} \\ \\ \\ \end{matrix}$$

$$\begin{pmatrix} 1 & 4 & 5 & 6 & 9 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 4 & 4 & 4 & 8 \\ 0 & -5 & -5 & -5 & -10 \end{pmatrix} \begin{matrix} \\ \\ R_3' = 4R_2 + R_3 \\ R_4' = R_4 - 5R_2 \end{matrix}$$

$$\begin{pmatrix} 1 & 4 & 5 & 6 & 9 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Basis for $R(A) = \{(1,4,5,6,9), (0,1,1,1,2)\}$

$\dim R(A) = 2$

Basis for $C(A) = \left\{ \begin{pmatrix}1\\0\\0\\0\end{pmatrix}, \begin{pmatrix}4\\1\\0\\0\end{pmatrix} \right\}$

$\dim C(A) = 2$

$\text{rank}(A) = \dim(R(A)) = \dim(C(A)) = 2$

Now,

To find the nullity of A, we must find the dimension of the solution space of the system, $Ax = 0$

$$x_1 + 4x_2 + 5x_3 + 6x_4 + 9x_5 = 0$$
$$x_2 + x_3 + x_4 + 2x_5 = 0$$

Let, $x_3 = s$, $x_4 = t$ and $x_5 = r$

$x_2 = -s - t - 2r$

$x_1 = 4s + 4t + \cancel{8}t - 5s - 6t - 9r$

$= -s - 2t - \cancel{1}r$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} -s-2t-r \\ -s-t-2r \\ s \\ t \\ r \end{pmatrix}$$

$$= s\begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t\begin{pmatrix} -2 \\ -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + r\begin{pmatrix} -1 \\ -2 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

So, the Nullspace of A,

$$N(A) = \left\{ s\begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t\begin{pmatrix} -2 \\ -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + r\begin{pmatrix} -1 \\ -2 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

Basis of $N(A) = \left\{ \begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ -2 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$

$\dim N(A) = 3$

Nullity $= 3$ Ans